

CS 1428: Foundations of Computer Science I

Fall 2025

Section R04

Instructor: Dr. Xiaomin Li
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Office Hours:
MW: 2:30pm – 4:30pm
in person in my office, zoom by request only

Learning Management System: Class materials will be available on **Canvas**.

Course Format (Mode of Delivery):
This is a **Face to Face** course.
MW 5:00 pm - 6:20 pm **Avery 00318**

Textbook:
We will be using an online textbook from **zyBooks**. It is required.

- Please see the Canvas Announcement on how to sign up for and subscribe to your zybooks book. When it asks for a student ID, **use your A0#!**
- If you opt out of BookSmart, the cost is \$89.
- For issues with zybooks: contact their support team: support@zybooks.com.

The lecture slides will also refer to a textbook: **Starting Out With C++ from Control Structures to Objects**, by Tony Gaddis (Editions 8, 9, and 10 are all fine)

Required In-Class Response system:
We will use the **Top Hat** classroom response system in class. You will be able to take attendance and submit answers to in-class questions using a mobile device or laptop. To register, go to the class Canvas site and click on Top Hat. When it asks for a student ID, **use your A0#!** Top Hat is integrated into Canvas and is free for Texas State students to use. There is a mobile app available, in addition to the Top Hat website.

Course Description:
Introductory course for computer science majors, minors and others desiring a technical introduction to computer science. The course emphasizes problem solving, algorithm development, structured programming, good coding style, and programming in C++.

Prerequisites: MATH 1315 or equivalent

Course Objectives:

At the end of the course, the students should be able to:

1. Describe the properties of good algorithms.
2. Design and develop good algorithms using a top-down approach.
3. Use the C++ programming language to implement, test, and debug algorithms for solving simple problems.
4. Explain the concepts of data types, variables, literals, and use them in programs.
5. Write C++ code that solves computational problems.
6. Use an if or if-else construct to implement branching in an algorithm.
7. Use a for loop for definite iteration.
8. Use a while or do-while loop for indefinite iteration.
9. Use functions and parameters to simplify longer programs and reuse code from previous solutions.
10. Demonstrate the mechanics of parameter passing with emphasis on the difference between pass by value and pass by reference.
11. Manipulate data in arrays.
12. Create a new data type by using a structure.
13. Analyze and explain the behavior of simple programs involving the fundamental programming constructs covered in this class.
14. Modify and expand short programs that use the constructs covered in this class.
15. Describe strategies that are useful in debugging.
16. Use a Windows- or Mac-based editor and compiler environment to develop programs in C++.

Course Schedule: There are 8 main topics or units:

Topic	
Unit 1	Intro to Programming & C++
Unit 2	Expressions & Data Types
Unit 3	Branches
Unit 4	Loops
Unit 5	File Input and Output
Unit 6	Functions
Unit 7	Arrays
Unit 8	Structs

Unit 9	Extra Topics (Optional)
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Exam coverage:

Midterm exam 1 covers Units 1-3.

Midterm exam 2 covers Units 4-6.

Final Exam covers all Units.

Grading:	Class Participation:	5%	Top Hat
	Weekly Homework	40%	zyBooks, Canvas
	Quizzes and In-class Practices	5%	Canvas
	Midterm Tests (2):	30%	On Paper
	Final Exam:	20%	On Paper

Final grade scales:

A: 100%~89.50%

B: 89.49%~79.50%

C: 79.49%~69.50%

D: 69.49%~59.50%

F: Below 59.49%

Class Participation:

Bring a web-enabled device to each class. You will be asked to take attendance using Top Hat. You can receive full credit for Class Participation if you attend more than 80% of the classes in person. You can receive half credit for Class Participation if you attend 50% ~ 80% of the classes in person. You will receive zero credit for Class Participation if you attend less than 50% of the classes in person.

Makeup Policy:

Make-up exams will be allowed only to students that were not able to take the original exam due to a health condition justified by the related paperwork from a doctor. Absence due to other reasons will be graded with zero.

Late Submission Policy:

Late assignments will incur a 10% penalty per day, for up to 3 days. After 3 days, no submission will be accepted. (If you have some special reasons, late assignment submissions are accepted till the end of the semester with a 40% penalty. You will need to ask for this special extension beforehand and I may deny your request.)

Examinations/Tests:

- All tests will be in class on paper, close-book, using pen or pencil, during class time.

Attendance policy:

Class attendance is very important, but it will not directly cause you to fail this course.

Notifications from the instructor:

Notifications related to this class will be posted in Canvas under Announcements (set up your Canvas notifications properly).

Use of AI Policy:

Definition. "AI" means large language models and LLM-powered tools (e.g., ChatGPT, Claude, Gemini, GitHub Copilot, CodeWhisperer) used to explain, draft, or generate text or code.

Allowed (limited use). You may use AI to:

- clarify concepts or terminology;
- brainstorm approaches or test ideas;
- view small illustrative code snippets for learning (do not submit them);
- get suggestions for debugging or fixing compile/runtime errors;
- polish grammar and wording of your own writing.

Not allowed. You may not:

- use AI to obtain or submit answers to any assignment, lab, or project;
- use AI to write any part of the code you submit;
- copy, adapt, or "heavily edit" AI-generated code or solutions and present them as your own;
- enable AI auto-completion/assistant extensions in your IDE while working on graded code (turn them off);
- use any AI tools during quizzes, tests, or exams.

Disclosure. If you consult AI for allowed purposes, briefly note it in your submission (tool name, prompt/task, how you used the output). Example: "Used ChatGPT to clarify xxxxx algorithm invariants; no code used."

Responsibility. You are fully responsible for the correctness, originality, and academic integrity of your work. You must be able to explain any submitted solution in detail.

Enforcement. Submitting AI-generated work (in whole or in part) or using prohibited AI during assessments is academic misconduct. Consequences may include a zero on the

assignment and referral under university policy.

Academic Honesty:

You are expected to adhere to the University's Academic Honor Code:

<https://www.txst.edu/honorcodecouncil/Academic-Integrity.html>

- All assignments must be done individually, with exceptions noted below.
- **Do not share your code electronically with anyone. Do not post your code on group chats or in email!**
- **Do not include code obtained from the internet or any other source in your programs.** Do not use AI tools to directly help you finish the homework. Do not use features we have not covered in class (without prior approval of the instructor).
- **Do no collaborate with anyone during a test or exam (no talking, no cell phones or smart watches).**
- The penalty for violating academic honesty on an assignment, test or exam is a reduction of points up to a grade of 0 on the assignment, test or exam.
- If you receive a 0 for cheating on an assignment, test or exam, I will report it to the Texas State University Honor Code Council (www.txst.edu/honorcodecouncil/).

Withdrawals/drops:

You must follow the withdrawal and drop policy set up by the University.

<https://www.registrar.txst.edu/resources/dropping-vs-withdrawing.html>

Accommodations for students with disability:

Texas State University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Services (DS) at (512) 245-3451, by email at ods@txstate.edu, or visit the [Disability Services web page](#).

Disabilities may include, but are not limited to: attentional, learning, mental health, sensory, physical, communication, or chronic medical conditions. All students are encouraged to discuss their disability related needs with the DS staff and their instructors as soon as possible.

Students who qualify for extra time for exams must take their exams with [ATSD](#) and must schedule their exam at the same time the exam is given in class. You must submit your request with ATSD at least 3 business days before the exam!